

Reproducible Regression Harness for Machine-Learning Security Tools

Contract-Validated Adapters Catching Cross-Repo Breakage Before It Ships

github.com/poojakira/mlsec-benchmark-suite (public, default branch main, primary language Python), MIT • Python 3.12 • HEAD 5c20f1da9602eb162efe8c63158da52b11d61f60 • verified 2026-10-01

EVIDENCE POSTURE

Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.

1.00

HF-SCANNER F1

3/0

TP / FP

83 across 10 modules

Test functions

82 passed, 1 skipped

Current CI

SYSTEM VIEW

Architecture and Threat-Control Flow

Trust Boundary

Input surface

Data, models, requests, policies, or artifacts entering the tested path.

Control surface

Repository-specific validation, analysis, and fail-closed checks.

Evidence surface

Findings, metrics, tests, logs, or reproducible artifacts.

This diagram describes the tested repository path, not a universal deployment guarantee.

01

Inspect the trust boundary

02

Apply repository-specific security checks

03

Record reproducible evidence

Research Contribution

A contract-validated regression harness that wraps ML security tools in typed in-process adapters, runs them against versioned fixtures, and fails pytest on schema drift - with a hard, labeled separation between REAL product adapters and a synthetic smoke-test scaffold.

Verified Evidence

EV-01 Statement coverage: 90.07%

EV-02 Real fixtures: 5

EV-03 Real adapters: gated CI

EV-04 Known-bad flagged: 100

EV-05 Known-good clean: 100

EV-06 Smoke scaffold (synthetic): 50

EV-07 VERIFIED_AT_SNAPSHOT - Real HF-scanner adapter; precision=recall=F1=1.0 on 5 fixtures - README verified real run; tests/test_hf_scanner_adapter.py end-to-end (no mock); total_tp=3,total_fp=0,total_fn=0.

EV-08 VERIFIED_AT_SNAPSHOT - 83 test functions across 10 test modules; current Python 3.12 CI recorded 82 passed, 1 skipped with 90.07% statement coverage - repository README and test tree at HEAD 5c20f1da9602eb162efe8c63158da52b11d61f60.

EV-09 VERIFIED_AT_SNAPSHOT - JSON-schema contracts + versioned fixtures; pytest fails on drift - README how-it-works; schemas/ + contracts/ present.

EV-10 UNSUPPORTED (disclaimed by repo) - run-smoke per-category numbers - README: seeded-PRNG synthetic scaffold for plumbing, NOT measurements. Shown as synthetic only.

Method and Control Path

01 Inspect the trust boundary

02 Apply repository-specific security checks

03 Record reproducible evidence

Limitations and Residual Risk

- Validates contracts/format, not real-world accuracy.
- run-smoke numbers are synthetic (seeded PRNG).
- Fixture coverage is finite; gaps possible.
- Adapters can go stale if a tool changes internals.
- Real adapter gating requires sibling install.

Reproducibility

Evidence: results/real_hf_scanner.json, METHODOLOGY.md, tests/

Evidence Provenance

Claim discipline

Metrics come from the repository's verified metrics and research brief.

Scope discipline

Benchmarks are reported only at their documented fixture, dataset, or test scope.

Reproduction

PDF and PNG are generated locally from committed evidence without paid services.